Engage+ Requirements Traceability Matrix

Purpose and scope

Engage+ is a mobile-first website for Howard students and organizations. This matrix traces 15 requirements to 25 features, 48 user stories, 48 backlog items and 59 tests, with no orphan requirements or features. Each feature has one owner from the five-person team.

The matrix is **bi-directional**. Forward, each requirement maps to features, then to stories, then to backlog items and tests. Backward, each feature maps to the requirement that justifies it.

Scope changes from the first draft

- Engage+ is a responsive website, not a native mobile app (PRD Non-Goals).
- R4 is now verified org accounts and event posting. It replaces the read-only Anthology Engage sync, because the PRD has orgs post their own events.
- Orgs post through a paste-box, with an opt-in browser extension that pulls events from Instagram.
- R10 is now calendar export. Safe commute moves to suggested features.

Source documents

- **SD-1 Problem statement:** Anthology Engage is desktop-first and search-driven, and has no mobile app for student discovery.
- **SD-2 Persona list:** 10 personas (P1–P10). The old P10 (night commuter) is replaced by a calendar-planner persona.
- **SD-3 Engage+ PRD and progress tracker:** 30 build steps, plus core features 1–21.

Team assignments

The 25 features are split by technical area so each person owns one connected part of the stack, with 4 to 6 features each. This split is a proposal for the team to adjust.

Member	Area	Features owned	Count

DM	AI and insights (Gemini)	F3, F5, F14, F16, F25	5
AD	Auth, org accounts and trust	F9, F11, F21, F24	4
PF	Student discovery front end	F1, F2, F4, F12, F13	5
SM	Org posting and extension	F8, F10, F15, F19, F22	5
JM	Social, notifications and engagement	F6, F7, F17, F18, F20, F23	6

Epics and user stories

Each persona is one epic, so there are 10 epics and 48 user stories. Stories use either "As a … I want … so that …" or "In order to … as a … I want …".

Epic 1 (P1, incoming first-year student): As an incoming first-year student who doesn't know any of Howard's 300+ organizations, I want an onboarding quiz that seeds a personalized swipe feed, so that I can find my community without knowing what to search for.

- **US-1.1:** As a first-year student, I want a short quiz on my interests, involvement type, major and year, so that the app knows what I care about from day one.
- **US-1.2:** In order to see the most relevant clubs first, as a first-year student I want my feed ranked by my quiz answers and swipe history.
- **US-1.3:** As a first-year student, I want to swipe right or left on event and org cards, so that I can RSVP, follow or skip in one gesture.
- **US-1.4:** As a brand-new user with no swipe history, I want to see a campus-wide trending feed, so that my feed is never empty.
- **US-1.5:** In order to get better recommendations over time, as a student I want to give a quick thumbs up or down after an event I RSVP'd to.
- **US-1.6:** As a student whose interests have changed, I want to retake the quiz, so that my feed stays relevant.

Epic 2 (P2, busy junior): As a junior balancing work and classes, I want targeted alerts from orgs I follow and the ability to mute specific orgs, so that I stop missing meetings without getting notification fatigue.

- **US-2.1:** As a busy junior, I want to follow an org by swiping right on its card, so that following takes one gesture.
- **US-2.2:** In order to manage my involvement in one place, as a busy junior I want a Following page that lists my followed orgs.
- **US-2.3:** As a busy junior, I want a notification when a followed org posts a new event, so that I don't have to check the site.
- **US-2.4:** In order to avoid notification fatigue, as a busy junior I want to mute and unmute individual orgs.
- **US-2.5:** As a student who rarely opens the site, I want a weekly "what's happening this week" digest, so that I can catch up without frequent alerts.

Epic 3 (P3, student org events chair): As an events chair, I want accurate RSVP counts before event day, so that I can order the right amount of catering and book the right-sized venue.

- **US-3.1:** In order to give organizers early turnout signals, as a student I want to RSVP by swiping right on an event.
- **US-3.2:** As an events chair, I want each student counted once per event, with RSVPs they can change, so that the counts I see are accurate.
- **US-3.3:** As an events chair, I want a per-event analytics page with live RSVP counts, so that I can plan food and space.
- **US-3.4:** In order to keep planning data inside our team, as an events chair I want only my org's managers to see our analytics.
- **US-3.5:** As an events chair, I want post-event thumbs up/down results in analytics, so that I can improve future events.

Epic 4 (P4, Office of Student Life administrator): As a Student Life administrator, I want every org account and event on Engage+ tied to a verified Howard student, so that listings are legitimate while Anthology Engage remains the official system of record.

- **US-4.1:** In order to keep the platform Howard-only, as an administrator I want sign-in through Microsoft restricted to Howard's tenant.
- **US-4.2:** As an administrator, I want org accounts creatable only by a verified student who enters the org's email, so that fake orgs are hard to create.
- **US-4.3:** As a founding student, I want to invite additional managers, including non-student staff such as an advisor, so that the people who run our org can manage its account.
- **US-4.4:** In order to get events onto Engage+, as an org manager I want to create and edit events with a flyer from an org dashboard.
- **US-4.5:** As an administrator, I want suspiciously fast follower growth flagged for manual review, so that inflated orgs get a human look.
- **US-4.6:** In order to understand campus engagement, as an administrator I want an aggregated, anonymized analytics view across orgs.

Epic 5 (P5, off-campus commuter): As a commuter on campus two days a week, I want events that clash with my classes hidden and the rest filtered by category, so that I only see activities that fit my gaps between classes.

- **US-5.1:** As a commuter, I want to upload my class schedule, so that the site knows when I'm busy.
- **US-5.2:** In order to see only events I can attend, as a commuter I want events that conflict with class time filtered out.
- **US-5.3:** As a commuter, I want to filter my swipe stack by category (academic, cultural, professional, social), so that I see the kind of events I want.
- **US-5.4:** As a returning student, I want a prompt to re-upload my schedule each semester, so that conflict filtering stays accurate.

Epic 6 (P6, org PR and social media chair): As a PR chair who already makes Instagram flyers, I want my captions turned into event listings automatically, so that I stop re-entering the same details into web forms.

- **US-6.1:** As a PR chair, I want to paste an Instagram caption and have AI fill in the title, date, time, location and category, so that posting takes seconds.
- **US-6.2:** As a PR chair, I want to upload the flyer image with the event, so that the listing looks like our Instagram post.
- **US-6.3:** In order to catch parsing mistakes, as a PR chair I want to review and edit the parsed fields before publishing.
- **US-6.4:** As a PR chair who has already posted via paste-box, I want a browser extension that shares an Instagram post to Engage+ in one click, so that I can skip the image upload too.
- **US-6.5:** In order to trust the extension, as a PR chair I want a plain-language explanation of exactly what it reads.
- **US-6.6:** As a PR chair, I want the extension to submit whatever fields it found when one is missing, so that I can fill the gap instead of starting over.
- **US-6.7:** As a PR chair, I want to set a recurrence rule for weekly meetings, so that I don't repost the same event every week.

Epic 7 (P7, student with a spontaneous break): As a student with a 90-minute break, I want to ask in plain language for something to do right now, so that I can find a nearby activity without digging through menus.

- **US-7.1:** As a student with a break, I want to type a question like "free cultural events this afternoon near the student union", so that I get answers without setting filters.
- **US-7.2:** In order to choose quickly, as a student I want each result to come with a one-line reason it matches.
- **US-7.3:** As a student, I want time words like "now" and "this afternoon" understood, so that results fit my actual break.
- **US-7.4:** As a student, I want to RSVP straight from a chat result, so that I don't have to find the event again.

Epic 8 (P8, social student): As a student who prefers going with friends, I want to see which friends have RSVP'd on event cards, so that I stop asking around in group chats.

- **US-8.1:** As a social student, I want to send and accept mutual friend connections, so that only people I approve see my RSVPs.
- **US-8.2:** In order to decide quickly, as a social student I want event cards in my normal feed to show connected friends who RSVP'd.
- **US-8.3:** As a private student, I want one setting that hides all my RSVPs from friends, so that I control my visibility.
- **US-8.4:** As a social student, I want to earn a verification badge by referring 3 active friends, so that bringing friends on is rewarded.

Epic 9 (P9, co-organizers of a joint panel): As co-organizers of a joint campus event, we want one event post credited to both orgs, so that updates reach both follower bases and turnout credit is shared.

- **US-9.1:** As a co-organizer, I want to tag multiple orgs on one event, so that every host gets credit.
- **US-9.2:** In order to reach both audiences, as a co-organizer I want the event shown on each co-host's page and sent to each co-host's followers.
- **US-9.3:** As the primary org, I want to keep edit and delete rights on a co-hosted event, so that there are no multi-owner conflicts.
- **US-9.4:** As a co-host, I want the event's RSVP numbers in my own analytics, so that turnout credit is shared.

Epic 10 (P10, calendar-planner student): As a student who plans my week in Google, Apple or Outlook Calendar, I want to export events I RSVP'd to, so that I don't forget them once I leave Engage+.

- **US-10.1:** As a planner, I want a Calendar page listing every event I RSVP'd to, so that I can see my week at a glance.
- **US-10.2:** In order to keep one calendar, as a planner I want to download my RSVP'd events as an .ics file.
- **US-10.3:** As a planner, I want the export to work in Google, Apple and Outlook without signing in to them, so that setup takes no extra accounts.

Requirements register

R1–R10 come from the 10 persona epics. R11–R15 are supporting requirements from the PRD that the epics depend on or extend. Priority uses MoSCoW: Must is in PRD steps 1–24, and Could is in appended steps 25–30.

Req ID	Requirement description	Source	User stories	Features	PRD steps	Priority
R1	Students must get a personalized swipe feed seeded by an onboarding quiz, refined by swipes and post-event feedback, with a trending fallback.	P1, PRD 1/5/10/11/13	US-1.1 to US-1.6	F1, F2, F3, F4, F5	5, 12, 14, 15, 17	Must
R2	Students must be able to follow orgs, get notified of their new events, and mute or unmute each org.	P2, PRD 2	US-2.1 to US-2.4	F2, F6, F7	5, 7	Must
R3	Students must be able to RSVP by swiping, and org managers must see per-event RSVP counts and feedback before and after the event.	P3, PRD 7	US-3.1, 3.2, 3.3, 3.5	F2, F5, F8	5, 6	Must
R4	Org accounts must be created only by verified students with an org email, support invited managers, and let managers post events with flyers.	P4, PRD 8/15	US-4.2 to US-4.5	F8, F9, F10, F11	3, 6, 11, 20	Must
R5	Students must be able to upload a class schedule that hides conflicting	P5, PRD 3/4	US-5.1 to US-5.3	F12, F13	8, 9	Must

	events and filter the stack by category.					
R6	Org members must be able to turn an Instagram caption into a structured event via AI, through the paste-box first and the opt-in extension later.	P6, PRD 9	US-6.1 to US-6.6	F10, F14, F15	10, 11, 19	Must (extension: Should)
R7	Students should be able to search events in natural language through an AI chat assistant.	P7, PRD 6	US-7.1 to US-7.4	F2, F16	13	Should
R8	Students should be able to make mutual friend connections, see friends' RSVPs on event cards, and hide their own RSVPs globally.	P8, PRD 14/14a	US-8.1 to US-8.3	F17, F18	18	Should
R9	Events should support multiple co-hosting orgs, with one primary owner and visibility to every co-host's followers.	P9, PRD 17	US-9.1 to US-9.4	F7, F8, F19	26	Should
R10	Students should be able to export RSVP'd events as an .ics file for Google, Apple or Outlook.	P10, PRD 12	US-10.1 to US-10.3	F20	16	Should

R11	All users must sign in with Microsoft OAuth restricted to Howard's tenant, with student, org and reserved admin roles.	P4, PRD 8, Tech Stack	US-4.1, US-3.4	F9, F21	3	Must
R12	Orgs could set a recurrence rule that generates future event occurrences.	P6, PRD 18	US-6.7	F22	27	Could
R13	The system could send a weekly digest and a schedule re-upload prompt each semester.	P2, P5, PRD 19	US-2.5, US-5.4	F23	28	Could
R14	Students could earn a verification badge after 3 referrals who each complete OAuth and a swipe session of 10+ swipes.	P8, PRD 16	US-8.4	F24	25	Could
R15	Student Activities staff could view aggregated, anonymized engagement analytics across orgs.	P4, PRD 20	US-4.6	F25	29	Could

Suggested features (not in this build)

These three features are recorded for traceability but have no backlog items or owners. They become requirements only if the team approves them and they are added to the PRD tracker.

ID	Suggested feature	Source	Why it is deferred
S1	Safe commute: well-lit route suggestions, hazard reports, and live location sharing with trusted contacts after evening events	Old P10 (urban commuter at night)	Outside Engage+'s core job of event and org discovery; needs mapping and real-time location services
S2	Expanded friend features: group-swiping, shared stacks, RSVP suggestions from friend activity	PRD 21, step 30	Builds on R8 only after base friend connections have real usage
S3	Sync RSVPs or events with Anthology Engage	First-draft R4, PRD Non-Goals	No official integration yet; would need Howard and Anthology approval

Feature and function register (backward traceability)

Each of the 25 features traces back to at least one requirement and to a PRD tracker step. F2, F7, F8 and F9 each serve several requirements.

Feature ID	Feature / function	Type	Description	Page or endpoint	PRD step	Traces to	Owner
F1	Org Discovery Quiz	Feature	Onboarding quiz on interests, involvement type, major and year	/onboarding, POST /users/quiz	14	R1	PF

F2	Swipe Discovery Feed	Feature	Swipe event cards to RSVP or pass, and org cards to follow or skip	/feed, POST /events/:id/swipe	5	R1, R2, R3, R7	PF
F3	AI Recommendation Engine	Function	Ranks the feed using quiz answers, then swipe history and feedback (Gemini)	GET /events/feed	12	R1	DM
F4	Trending Feed	Feature	Campus-wide, most RSVP'd and followed this week; fallback for new users	GET /events/trending	15	R1	PF
F5	Post-Event Feedback	Function	Thumbs up/down prompt after an RSVP'd event passes	/feedback/:eventId	17	R1, R3	DM
F6	Follow and Per-Org Mute	Feature	Following list with mute and unmute	/following, POST /orgs/:id/mute	7	R2	JM

			toggles per org				
F7	Notification Service	Function	Alerts followers (including all co-hosts' followers) when an org posts an event	Backend job	7	R2, R9	JM
F8	Org Dashboard and Analytics	Feature	Event list, per-event RSVP and feedback stats, manager management	/org/dashboard, /org/analytics/:eventId	6	R3, R4, R9	SM
F9	Org Account Creation Gate and Manager Invites	Function	Only verified students create orgs (with org email); founder invites managers, including staff	/org/create, POST /orgs/:id/managers/invite	3	R4, R11	AD
F10	Event Posting Form and Flyer Upload	Feature	Paste-box form, flyer upload to Supabase Storage,	/org/new-event, POST /events	11	R4, R6	SM

			review and publish				
F11	Follower-Velocity Flag	Function	Flags suspiciously fast Instagram follower growth for manual review	Backend job	20	R4	AD
F12	Class Schedule Upload and Conflict Filter	Feature	Student enters class blocks; conflicting events are hidden from the feed	POST /users/schedule	8	R5	PF
F13	Category and Interest Filters	Feature	Filter the swipe stack by academic, cultural, professional, social and other categories	/feed	9	R5	PF
F14	AI Caption Parser	Function	Gemini extracts title, date, time, location and category	POST /events/parse-caption	10	R6	DM

			from caption text				
F15	Browser Extension (Manifest V3)	Feature	Opt-in, instagram.com-only extension that sends caption, image and follower count	POST /events/extension-submit	19	R6	SM
F16	AI Chat Assistant	Feature	Natural-language event search with a one-line reason per result	/chat	13	R7	DM
F17	Friend Connections and Friends' RSVPs	Feature	Mutual, unlimited connections; event cards show friends who RSVP'd	/friends, GET /events/:id/friends-going	18	R8	JM
F18	Hide My RSVPs Toggle	Function	One global setting that hides your RSVPs from all friends	/settings/privacy, POST /users/hide-rsvps	18	R8	JM

F19	Co-Hosted Events (EventCohost)	Feature	Tag several orgs on one event; the primary org keeps edit and delete rights	EventCohost table	26	R9	SM
F20	Calendar Export (.ics)	Feature	Calendar page of RSVP'd events with .ics download	/calendar, GET /events/export.ics	16	R10	JM
F21	Microsoft OAuth and Role-Based Access	Function	Sign in with Microsoft, restricted to Howard's tenant; student, org and reserved admin roles	/login, GET /auth/microsoft	3	R11	AD
F22	Recurring Events (EventRecurrence)	Feature	Recurrence rule that generates real Event rows about 8 weeks ahead	EventRecurrence table	27	R12	SM

F23	Weekly Digest and Semester Nudge	Function	Weekly "what's happening" summary and a schedule re-upload prompt each semester	Backend job	28	R13	JM
F24	Referral Flow and Verification Badge	Feature	Badge after 3 qualifying referrals (OAuth plus 10+ swipes each)	Referral table	25	R14	AD
F25	Student Activities Admin View	Feature	Aggregated, anonymized turnout, follow and category trends	Admin role	29	R15	DM

Traceability matrix (requirements × features)

Each filled cell holds the initials of the member who builds that feature for that requirement, as in the course's Example 3. Every row and every column has at least one entry. The matrix is split into two parts to fit the page.

Part A: features F1–F13

Requirement	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13

R1. Personalized swipe feed	PF	PF	DM	PF	DM									
R2. Follow, notify, per-org mute		PF				JM	JM							
R3. RSVP counts for organizers		PF			DM			SM						
R4. Verified org accounts and posting								SM	AD	SM	AD			
R5. Schedule and category filters												PF	PF	
R6. AI caption to event										SM				

| | | |

R14. Referral badge

R15. Admin analytics view

Part B: features F14–F25

Requirement	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25
R1. Personalized swipe feed												
R2. Follow, notify, per-org mute												
R3. RSVP counts for organizers												

R4. Verified org accounts and posting									
R5. Schedule and category filters									
R6. AI caption to event	DM	SM							
R7. Natural-language search			DM						
R8. Friends' RSVPs				JM	JM				
R9. Co-hosted events						SM			
R10. Calendar export							JM		

R11. Howard OAuth and roles	AD	
R12. Recurring events	SM	
R13. Digest and semester nudge	JM	
R14. Referral badge	AD	
R15. Admin analytics view	DM	

Feature key: F1 Quiz · F2 Swipe Feed · F3 AI Recommendations · F4 Trending · F5 Feedback · F6 Follow and Mute · F7 Notifications · F8 Org Dashboard · F9 Org Gate and Managers · F10 Posting Form · F11 Velocity Flag · F12 Schedule Filter · F13 Category Filters · F14 Caption Parser · F15 Extension · F16 AI Chat · F17 Friends · F18 Hide RSVPs · F19 Co-Hosting · F20 Calendar Export · F21 Microsoft OAuth · F22 Recurring · F23 Digest · F24 Referrals · F25 Admin View.

Backlog items: Epics 1–5

Epics 1–5 break down into 26 backlog items. Each item maps to one user story and one acceptance test (TC-01 to TC-26). Backlog IDs follow the course format F#_##.

Backlog ID	Task	User story	Req	Owner	Acceptance test
F1_01	Build onboarding quiz (interests, involvement type, major, year)	US-1.1	R1	PF	TC-01: New user finishes the quiz and answers save through POST /users/quiz
F3_01	Rank feed by quiz answers, then by swipe history	US-1.2	R1	DM	TC-02: Two users with different answers get different top-5 events
F2_01	Build swipe card stack for event and org cards	US-1.3	R1	PF	TC-03: Right swipe on an event saves an RSVP; left saves a pass
F4_01	Show trending feed when no quiz or swipe data exists	US-1.4	R1	PF	TC-04: A user with no data sees trending events, not an empty feed
F5_01	Prompt thumbs up/down after an RSVP'd event ends	US-1.5	R1	DM	TC-05: Prompt appears after the end time and the rating saves
F1_02	Let students retake the quiz	US-1.6	R1	PF	TC-06: New answers change the feed on next load
F2_02	Follow an org by swiping right on its card	US-2.1	R2	PF	TC-07: A Follow record is created

F6_01	Build Following page	US-2.2	R2	JM	TC-08: Page lists exactly the followed orgs
F7_01	Notify followers when an org publishes an event	US-2.3	R2	JM	TC-09: Follower is notified after the event is published
F6_02	Add per-org mute and unmute toggle	US-2.4	R2	JM	TC-10: A muted org sends no alerts; unmuted orgs still do
F23_01	Generate weekly "what's happening" digest	US-2.5	R13	JM	TC-11: Digest lists the week's followed and trending events
F2_03	Record RSVP on right swipe of an event	US-3.1	R3	PF	TC-12: Event RSVP count rises by 1
F2_04	Allow one RSVP per student per event, changeable	US-3.2	R3	PF	TC-13: Repeat swipes never double-count; switching to pass lowers the count
F8_01	Build per-event analytics page with RSVP count	US-3.3	R3	SM	TC-14: Count matches stored Swipe records
F21_01	Restrict analytics to the org's managers	US-3.4	R11	AD	TC-15: A non-manager gets 403 from GET /orgs/:id/analytics

F8_02	Show post-event feedback totals in analytics	US-3.5	R3	SM	TC-16: Thumbs totals match EventFeedback records
F21_02	Sign in with Microsoft, restricted to Howard's tenant	US-4.1	R11	AD	TC-17: A non-Howard Microsoft account is rejected
F9_01	Gate org creation behind a student session and org email	US-4.2	R4	AD	TC-18: POST /orgs without a student session is rejected
F9_02	Invite additional managers, including non-student staff	US-4.3	R4	AD	TC-19: Invited staff member can post events; isStaff is true
F10_01	Build event create and edit form with flyer upload	US-4.4	R4	SM	TC-20: Published event with its flyer appears in the feed
F11_01	Flag fast follower growth for manual review	US-4.5	R4	AD	TC-21: A large jump in a short window sets the flag; nothing is blocked
F25_01	Build aggregated, anonymized admin analytics view	US-4.6	R15	DM	TC-22: Admin sees totals only; no individual student is identifiable
F12_01	Build class schedule upload	US-5.1	R5	PF	TC-23: Schedule blocks save through POST /users/schedule

F12_02	Hide events that conflict with class blocks	US-5.2	R5	PF	TC-24: An event overlapping a class is absent from the feed
F13_01	Add category filter to the swipe stack	US-5.3	R5	PF	TC-25: Only events in the selected category show
F23_02	Prompt schedule re-upload each semester	US-5.4	R13	JM	TC-26: Prompt appears at the first sign-in of a new semester

Backlog items: Epics 6–10

Epics 6–10 break down into 22 backlog items, mapped to user stories and to tests TC-27 to TC-48.

Backlog ID	Task	User story	Req	Owner	Acceptance test
F14_01	Parse pasted caption with Gemini into structured fields	US-6.1	R6	DM	TC-27: Sample captions return title, date, time, location and category as JSON; missing fields are null
F10_02	Upload flyer image to Supabase Storage	US-6.2	R6	SM	TC-28: Image is stored and its URL is saved on the event
F10_03	Review and edit parsed fields before publishing	US-6.3	R6	SM	TC-29: Every field is editable; nothing goes

					live until the member confirms
F15_01	Build "Share to Engage+" extension, offered after the first paste-box post	US-6.4	R6	SM	TC-30: Extension is offered only after 1 paste-box post; sharing creates a draft
F15_02	Write plain-language permission screen; limit scope to instagram.com	US-6.5	R6	SM	TC-31: Manifest has no all-URLs permission; install screen shows the explanation
F15_03	Submit partial data when a field isn't found	US-6.6	R6	SM	TC-32: With follower count missing, caption and image still submit
F22_01	Generate occurrences from a recurrence rule	US-6.7	R12	SM	TC-33: A weekly rule creates Event rows about 8 weeks ahead
F16_01	Build chat page that accepts free-text questions	US-7.1	R7	DM	TC-34: A typed question returns matching upcoming events
F16_02	Return a one-line reason with each result	US-7.2	R7	DM	TC-35: Every result shows a reason
F16_03	Interpret time words like "now" and "this afternoon"	US-7.3	R7	DM	TC-36: "This afternoon" returns only same-day afternoon events

F2_05	RSVP directly from a chat result	US-7.4	R7	PF	TC-37: RSVP from chat appears in the student's Swipe records
F17_01	Send and accept mutual friend requests, with no cap	US-8.1	R8	JM	TC-38: Connection stays pending until accepted
F17_02	Show friends who RSVP'd on feed event cards	US-8.2	R8	JM	TC-39: Card lists only accepted friends who RSVP'd
F18_01	Add global "Hide my RSVPs from friends" toggle	US-8.3	R8	JM	TC-40: A hidden user is absent from friends' cards but still sees friends' RSVPs
F24_01	Track referrals and award badge at 3 qualifying referrals	US-8.4	R14	AD	TC-41: Badge is awarded only when 3 referrals each completed OAuth and 10+ swipes
F19_01	Tag multiple orgs on one event (EventCohost)	US-9.1	R9	SM	TC-42: Event stores 2 or more co-host orgs
F7_02	Show event on each co-host's page and notify all co-hosts' followers once	US-9.2	R9	JM	TC-43: A student following both hosts gets 1 alert, not 2

F19_02	Limit edit and delete rights to the primary org	US-9.3	R9	SM	TC-44: A co-host manager's edit attempt is rejected
F8_03	Include co-hosted RSVPs in each co-host's analytics	US-9.4	R9	SM	TC-45: Both orgs' analytics show the same RSVP count
F20_01	Build Calendar page of RSVP'd events	US-10.1	R10	JM	TC-46: Page lists every RSVP'd upcoming event
F20_02	Generate .ics file from RSVP'd events	US-10.2	R10	JM	TC-47: File validates and contains every RSVP'd event
F20_03	Confirm import works in major calendar apps	US-10.3	R10	JM	TC-48: File imports into Google, Apple and Outlook with no sign-in

Non-functional and technical requirements

The 5 non-functional requirements constrain every feature. The 8 technical requirements cover the PRD build steps that deliver no user-facing feature (steps 1, 2, 4 and 21–24). Each has an owner and a check.

ID	Requirement	Source	Affects	Owner	Verified by
NFR1	Engage+ is a mobile-first, responsive website; no native iOS or Android apps.	PRD Overview, Non-Goals	All	PF	TC-49: Core flows work at phone (375 px) and desktop widths in

					Chrome and Safari
NFR2	Howard-only: sign-in restricted to Howard's Microsoft 365 tenant; no multi-university support.	PRD Out of Scope	F21	AD	TC-17
NFR3	Privacy: analytics only for org managers, admin data aggregated and anonymized, RSVPs hideable from friends.	PRD 8, 14a, 20	F8, F18, F25	AD	TC-15, TC-22, TC-40
NFR4	Extension resilience: isolated DOM-reading module, scheduled selector check, paste-box kept as a permanent fallback.	PRD Extension Rollout	F10, F15	SM	TC-50: Monitoring job alerts when selectors fail on the known test post
NFR5	Run at $0 on free tiers, plus a $5 one-time Chrome Web Store fee.	PRD Tech Stack	All	DM	TC-51: Every hosted service is on a free plan at deployment
TR1	Project setup: React + TypeScript (Vite), Express + TypeScript, Prisma, Tailwind	Step 1	All	DM	TC-52: Front end and back end build and run locally
TR2	Docker and Docker Compose	Step 2	All	AD	TC-53: One compose command starts front end, back end and database

TR3	Data models: User, Org, OrgManager, Event, Swipe, Follow, ScheduleBlock, EventFeedback, Friendship, Referral, EventCohost, EventRecurrence	Step 4	All	AD	TC-54: Prisma schema migrates with all 12 models
TR4	Testing with Jest (unit) and Playwright (end-to-end)	Step 21	All	PF	TC-55: Both suites run and pass, including TC-01 to TC-48
TR5	CI/CD with GitHub Actions	Step 22	All	AD	TC-56: Every pull request triggers build and tests
TR6	Deployment: Vercel (front end), Render (back end), Supabase (database and storage)	Step 23	All	DM	TC-57: Production site loads and reads from Supabase
TR7	Polish: responsive layout, loading states, error handling	Step 24	All	PF	TC-58: Every page shows a loading and an error state
TR8	Code standards: TypeScript only, no file over 300 lines	PRD rules	All	All members	TC-59: Lint check fails any file over 300 lines

Coverage summary and gaps

Coverage is complete in both directions: 15 requirements, 25 features, 48 user stories, 48 backlog items and 59 tests, with no orphans. Five open decisions remain, listed below the table.

Req	Features	User stories	Backlog items	Tests	Owners
R1	5	6	6	TC-01 to 06	PF, DM
R2	3	4	4	TC-07 to 10	PF, JM
R3	3	4	4	TC-12 to 14, TC-16	PF, SM
R4	4	4	4	TC-18 to 21	AD, SM
R5	2	3	3	TC-23 to 25	PF
R6	3	6	6	TC-27 to 32	DM, SM
R7	2	4	4	TC-34 to 37	DM, PF
R8	2	3	3	TC-38 to 40	JM
R9	3	4	4	TC-42 to 45	SM, JM
R10	1	3	3	TC-46 to 48	JM
R11	2	2	2	TC-15, TC-17	AD

R12	1	1	1	TC-33	SM
R13	1	2	2	TC-11, TC-26	JM
R14	1	1	1	TC-41	AD
R15	1	1	1	TC-22	DM

Backlog items per member: JM 12, SM 12, PF 11, DM 7, AD 6. DM's and AD's items are fewer but heavier (Gemini integration, OAuth), and they also own 8 of the 13 non-functional and technical requirements.

Open decisions

- Choose the notification channel for a website: web push, email, or both (F7, F23).
- Confirm the new P10 calendar-planner persona with the team, since it replaces the night-commuter persona.
- Decide whether to keep the first draft's time and location filters; the PRD has only category and schedule filters.
- Define when a semester starts so TC-26 has a fixed trigger date.
- Scope the admin view (F25) with a Howard Student Activities contact before building, as the PRD requires.
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